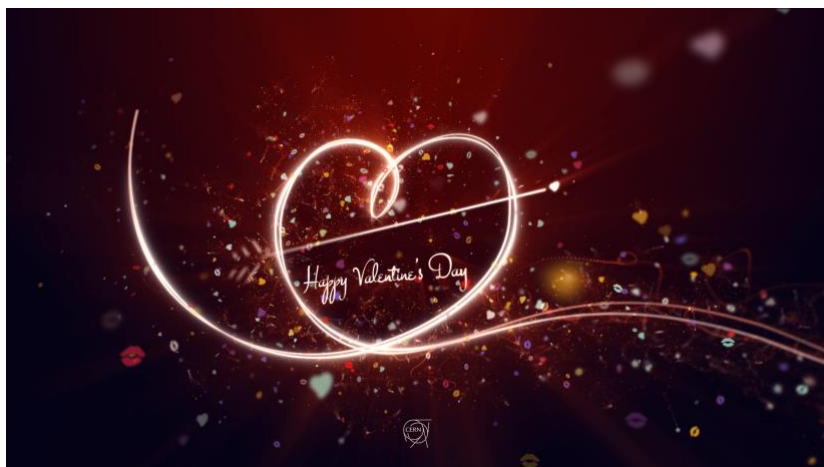


Quadrupoles are red, dipoles are blue...



CERN community: this Valentine's Day we're asking you to compose an ode to technology.

Send us your CERN-related Valentine's poem, written in English or French, and we'll publish our favourites in the next Bulletin. We'll also give a prize to the poem that we like the best. Poems must be a maximum of 20 lines, and the more CERN-specific the better.

Here are our attempts to get you started:

*"Quadrupoles are red
Dipoles are blue
CERN's magnets are cool
They're attractive too"*

*"Protons are red,
Electrons are blue,
There's neutrons as well
But you are my glu(on)"*

*"There once was an experimentalist,
Who was a great friend of a theorist.
She said to him:
"I give thee this ring"
And LHC made them their happiest."*

Send your poem to bulletin-editors@cern.ch by **midnight CET on Sunday, 25 February**. Please note that you must have a CERN email address to enter.

By taking part in this competition, you accept that your poem may be published in the next CERN Bulletin. If you wish, you can request that we publish it anonymously.

A Word from Mike Lamont

Chamonix: building on success, planning for the future

From 29 January to 1 February, the Chamonix workshop 2024 upheld its long tradition of fostering open and collaborative discussions within CERN's accelerator and physics communities

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A Word from Mike Lamont

Chamonix: building on success, planning for the future

*Mais où sont les neiges d'antan?** From 29 January to 1 February, the Chamonix workshop 2024 upheld its long tradition of fostering open and collaborative discussions within CERN's accelerator and physics communities. This year marked a significant shift with more explicit inclusion of the injector complex, acknowledging its crucial role in shaping future research endeavours. Chamonix discussions focused on three main areas: maximising the remaining years of Run 3; the High-Luminosity LHC (HL-LHC), preparations for Long Shutdown 3 and operations in Run 4; and a look to the further future and the proposed Future Circular Collider (FCC).

Analysing the **performance** of CERN's accelerator complex, speakers noted the impressive progress to date, examined limitations in the LHC and injectors and discussed improvements for optimal performance in upcoming runs. It's difficult to do justice to the immense technical effort made by all systems, operations, and technical infrastructure teams that underpins the exploitation of the complex. Machine **availability** emerged as a crucial theme, recognised as critical for both maximising the potential of existing facilities and ensuring the success of the HL-LHC. Fault tracking, dedicated maintenance efforts, and targeted infrastructure improvements across the complex were highlighted as key contributors to achieving and maintaining optimal uptime.

As the **HL-LHC** project moves into full series production, the technical challenges associated with magnets, cold powering and crab cavities are being addressed. Looking beyond **Long Shutdown 3 (LS3)**, potential limitations are already being targeted now, with, for example, electron cloud mitigation measures planned to be deployed in LS3. The transition to the high-luminosity era will involve a huge programme of work that requires meticulous preparation and a well-coordinated effort across the complex during LS3, which will see the deployment of HL-LHC, a widespread consolidation effort and other upgrades such as that planned for the ECN3 cavern at CERN's North Area.

The breadth and depth of the physics being performed at **CERN facilities** is quite remarkable, and the Chamonix workshop reconfirmed the high demand from experimentalists across the board. The unique capabilities of ISOLDE, n_TOF, AD-ELENA and the East and North Areas were recognised. The North Area, for example, provides protons, hadrons, electrons and ion beams for detector R&D, experiments, the CERN neutrino platform and irradiation facilities, and counts more than 2000 users.

The vision for the next decades of these facilities is diverse, imaginative and well-motivated from a physics perspective. The potential for long-term exploitation and leveraging fully the capabilities of the LHC and other facilities is considerable, demanding continued support and development. In the longer term, CERN is exploring the potential construction of the **FCC** via a dedicated feasibility study that has just delivered a mid-term report – a summary of which was presented at Chamonix and was presented to the CERN community on [13 February](#). The initiative is accompanied by R&D on key accelerator technologies. The physics case for FCC-ee was well made for an audience of mostly non-particle physicists, concluding that FCC is the only proposed collider that covers each key area in the field – electroweak, QCD, flavour, Higgs and searches for phenomena beyond the Standard Model – in paradigm-shifting depth.

Sustainability was another focus of the Chamonix workshop. Building and operating future facilities with environmental consciousness is a top priority, and full life-cycle analyses will be performed for any options to help ensure a low-carbon future. Interesting times, lots to do. To quote former CERN Director-General [Herwig Schopper](#) from 1983: “it is therefore clear that, for some time to come, there will be interesting work to do and I doubt whether accelerator experts will find themselves without a job”.

** Extract from the poem “Ballade des dames du temps jadis” by François Villon.*

Mike Lamont
Director for Accelerators and Technology

Accelerator Report: The accelerator complex gears up for action after the yearly winter maintenance break

The symbolic key to resume LHC operations will be handed over from the ACE (Accelerator Coordination and Engineering) group in the Engineering department to the Operations group on Friday, 16 February, kicking off the 2024 “particle season”.

As winter bids farewell, the recommissioning of the accelerator complex gathers pace, with the scientific community eagerly awaiting particle beams in their experiments. Following the year-end technical stop (YETS), Linac4 is the first machine to resume beam operation, followed by the downstream machines: the PS Booster, PS, SPS and LHC.

Beam entered Linac4 on 5 February, two days ahead of schedule – extra time welcomed by the Linac team. During the YETS, work was done on the chain of accelerating cavities, requiring a re-phasing – a challenging and often time-consuming task. To do so, the acceleration of the particle beam is optimised as the beam goes down the Linac: the voltage waves in the cavities are timed correctly as the beam passes by, ensuring optimum acceleration in each of the cavities and bringing the energy to 160 MeV at the end of the Linac.

This week, the beam was then sent to the PS Booster. The operations team has one week to prepare for the first beam to be injected into the PS on 21 February. The PS will then have to prepare the first beam for the SPS beam commissioning, scheduled to start on 1 March. The first particle beams will reach the LHC on 11 March, initially with one to a few bunches at most. Before injecting particle beams, the hardware recommissioning coordinators of each machine and the many equipment experts have the task of meticulously recommissioning and validating all the subsystems. They run the machine “as if” particle beams were being accelerated, but without particles. They go through checklists, validating and ticking off thousands of tests, to give the green light for beam commissioning.

The expectations for 2024 are high. Firstly, in the LHC, the focus is on luminosity production with proton–proton collisions, aiming at an

unprecedented accumulation of luminosity of up to 90 fb^{-1} . This, together with the accumulation of luminosity forecast for the 2025 run, should provide a sizeable analysis data set to keep physicists busy during Long Shutdown 3. The 2024 LHC run will conclude with lead–lead collisions; the first lead ions will be injected into the LHC on 6 October. The 2024 run is scheduled to end on 28 October.

The injector chain has an ambitious year ahead as well: the injectors have a busy fixed-target programme and will provide beams to all the experimental facilities. The first fixed-target physics will start in the PS East Area on 22 March, followed by the PS n_TOF facility on 25 March. Physics in ISOLDE, downstream of the PS Booster, will start on 8 April, followed by the SPS North Area on 10 April. The antimatter factory is set to start delivering antiprotons to its experiments on 22 April. The AWAKE facility, behind the SPS, will run for ten weeks in total (in blocks of two or three weeks) until the middle of September, when the dismantling of the no-longer-used CERN Neutrinos to Gran Sasso (CNGS) target facility will start, to allow for a future extension of the AWAKE facility. The SPS HiRadMat facility will see four 1-week runs.

Beyond this busy physics programme, many machine development studies and tests are planned in all the machines. One of these tests will take place between mid-March and early June to configure the Linac3 source to produce magnesium ions, which will be accelerated in Linac3, injected into LEIR, and possibly even into the PS. This test will help assess the feasibility and performance of magnesium beams in the accelerator complex, for potential future applications in the LHC and the SPS North Area.

The resumption of operation of the accelerator complex heralds a new year of physics, surely leading to important physics results. As the countdown to 11 March continues, the operations and expert teams are working diligently to prepare the machines and the beams for another successful physics run.

Rende Steerenberg

CERN70: A first discovery

Part 1 of the [CERN70](#) feature series: Giuseppe Fidecaro was among the small group of physicists who performed the first experiment at CERN to provide results in 1958 that would spread the Laboratory's name around the world



The 600 MeV Synchrocyclotron (SC) was CERN's first accelerator. It began operation in 1957 and ran for 33 years. The SC is now one of the visit points for CERN guided tours. (Image: CERN)

A few months after CERN's first accelerator, the Synchrocyclotron (SC), was commissioned, a first experiment was launched. At the time, weak interactions were among the most hotly debated topics in high-energy physics. Scientists were puzzled, for example, about the decay of the particle known as the pion. The particle was known to decay into two other particles: a muon and a neutrino. According to theory, it should also sometimes decay into an electron and a neutrino, but this type of decay had never been observed before. In August 1958, at CERN's Synchrocyclotron, Tito Fazzini, Giuseppe Fidecaro, Alec Merrison, Helmut Paul and Alvin Tollestrup observed this decay for the first time, at a rate in line with predictions of the theory of the weak interaction.

It was CERN's first major discovery.

Recollections

Giuseppe Fidecaro arrived in Geneva in 1956, after spending two years at Liverpool University's synchrocyclotron. He was one of the small group of physicists who performed the first experiment at CERN to provide results that would spread the Laboratory's name around the world.



The Synchrocyclotron (SC) building with the Jura mountains in the background, photographed in December 1958. (Image: CERN)

“The basic idea of the experiment was that the beam of pions would be stopped in a scintillation counter. Our experimental set-up consisted of a telescope made of a sandwich of graphite plates and scintillation counters. The output signals were displayed on the screen of a high-speed oscilloscope and photographed on continuously moving film. We could recognise the decay of the pion from the recorded signal. With its twelve counters, it was a rather complex and sophisticated apparatus for its time!

In August 1958, we finally got beam to our apparatus. After some checks, we went on changing the thickness of the telescope. A thin telescope simply gave muon and neutrino decay signatures. When we inserted the last graphite plate, we weren't really prepared for what was going to happen. Tito Fazzini, Alec Merrison, Helmut Paul, Alvin Tollestrup and I observed what no one else in the world had seen before: the film essentially showed electron and neutrino decay signatures. We had pion-electron events! A few days later, Julius Ashkin joined our experiment.

On 1 September, the second United Nations International Conference on the Peaceful Uses of Atomic Energy opened its doors in Geneva. Our results couldn't be hidden from the large number of physicists, journalists etc. who had come from all over the world for the conference. Although the experiment was not yet complete, we decided to present what we already had – 40 events, from a total of 124 photographs, at a rate compatible with theory – at a special session of the conference on Thursday 4 September, only one month after we got the first beam. The next day, the news had

gone around the world. For the first time the local newspaper *Tribune de Genève* announced “Discovery at CERN”. Our first paper was accepted for publication one week later, and the final paper appeared in July 1959.



Giuseppe and Maria Fidecaro, pioneers of CERN's experiments, became key figures in the Laboratory, continuing to work until very recently. Read the [CERN Courier tribute](#) to Maria, who passed away last September. (Image: CERN)

Our approach had been purely experimental, typical of a university lab. An old problem had suddenly turned hot and we looked at whether it could be solved with existing equipment. An experiment using a range telescope had failed to find the missing decay a few years earlier. We could repeat it – but with a minor modification: pions would stop in a scintillator, not within an inert target. A so-called travelling wave oscilloscope was already in our hands! Marc Fell and Max Renevey only had to build a few light-guides and mechanical supports. All this took place in an atmosphere full of enthusiasm.

CERN hosted the Rochester Conference [now ICHEP (International Conference on High Energy Physics)] in June 1958, less than two months after our work had begun. Our apparatus was set up in the “neutron hall” of the SC, and the conference participants were able to see it. To some of them, the telescope looked quite simplistic, compared to the enormous magnet used for the same purpose in Chicago the previous year, but without success.

There were additional firsts in this experiment. The programme written for CERN's brand-new Mercury computer, designed to determine the efficiency of our apparatus, was the first "Monte Carlo" programme – based on a randomised simulation – written for an experiment at CERN. Together with Yves Goldschmidt-Clermont and Norman Lipman, we were also able to measure the average lifetime of the pion with improved precision. To do this, we used the first instrument

developed at CERN to evaluate photos, IEP, which, while we waited for the bubble chamber images, enabled us to evaluate the 10 000 events recorded with our apparatus".

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary.

CERN Council reviews progress of feasibility study for a next-generation collider

At its half-way mark, a study investigating the feasibility of a 91-kilometre Future Circular Collider to potentially follow the Large Hadron Collider at CERN shows significant progress

After three years of work, mobilising the expertise of scientists and engineers around the world, the Feasibility Study for the Future Circular Collider (FCC) - a particle collider with a circumference of 90.7 km that could potentially succeed the High-Luminosity LHC in the mid-2040s – has now reached the half-way mark. The Feasibility Study is expected to be completed in 2025.

The CERN Council reviewed the work undertaken in a fruitful meeting on 2 February 2024. It congratulated and thanked all the teams involved in the study for the excellent and significant work done so far and for the impressive progress, and looks forward to receiving the final report in 2025. Particle colliders have played a crucial role in elucidating the fundamental laws of nature and constituents of matter. The Feasibility Study for the FCC was launched in response to a recommendation from the 2020 update of the European Strategy for Particle Physics, whereby Europe, in collaboration with the worldwide community, should undertake a technical and financial feasibility study for a next-generation hadron collider at the highest achievable energy, with an electron-positron collider as a possible first stage.

If approved by CERN's Member States in the coming years, the construction of the first stage, an electron-positron collider (FCC-ee), could start in the early 2030s and operate in the mid-2040s. The facility would operate for some 15 years, during which time the high-field magnet technology needed for the second stage, a proton-proton collider operating at an unprecedented collision energy of around 100 TeV (FCC-hh), could be developed and industrialised.

Accelerator, detector, and physics studies continue within the global FCC collaboration, spanning 150 institutes in 30 countries.

Relevant links:

- <https://home.cern/news/press-release/accelerators/international-collaboration-publishes-concept-design-post-lhc>
- <https://home.cern/news/press-release/cern/cern-prepares-its-long-term-future>
- [Status of the FCC Feasibility Study - recording of CERN community presentation](#)

Stories from CERN's women in science

The Organization's cooperation with the CERN celebrates the International Day of Women and Girls in Science by shedding light on the variety of career paths for women in STEM

Each year, on 11 February, CERN celebrates the International Day of Women and Girls in Science by shedding light on the variety of career paths for women in STEM. This year, we asked nine female scientists to share their stories with us and tell us what inspired them to pursue a STEM career and what are their favourite memories involving science.



Sorina, physicist at the CMS experiment

Sorina is a Romanian research physicist at the [CMS experiment](#), working on heavy-ion research.

"My favourite thing about my job is the data analysis, as well as detector development and installation. I am very happy when I can work with the students. The most rewarding part of it is when I see their careers as physicists evolve."

Jenny, PhD student at the ATLAS experiment

Jenny is a Norwegian doctoral student at the University of Oslo. She's currently working on upgrading the pixel detector for the [ATLAS experiment](#).

Science was her favourite class in school, which inspired her to pursue a career in STEM. "My curiosity for science started with what we can see in our everyday lives, for example how yeast makes bread dough rise, how a candle flame goes out if it loses access to oxygen, or how nature changes with the seasons."

Pinelopi, PhD student with the Medipix collaboration

Pinelopi is a PhD student with the [Medipix collaboration](#). Medipix chips developed for pixel

detectors at the LHC are now used in a variety of fields, including medical imaging.

It was her family and her secondary school experiences that inspired Pinelopi to pursue a career in STEM. "My mother studied physics, so I wanted to become like her, while my father loves to explore ideas and think outside the box. As a secondary school student, I visited CERN with my class, and I was amazed by everything. My dream was to one day return as a physicist."

Federica, PhD student at the LHCb experiment

Federica is an Italian doctoral student in particle physics, working on heavy-ion physics at the [LHCb experiment](#). She is currently involved in putting the VELO detector into service.

Federica has always been curious about science and knew from an early age that she was going to pursue a STEM career. Her favourite memory involving science goes back to secondary school, when two CERN physicists visited her class to give a masterclass. "They built a hand-made cloud chamber for us, with things you could find in the kitchen, to detect the particles from cosmic rays. And I fell in love with particle physics!" remembers Federica.

Joni, physicist at the ATLAS experiment

Joni, a Vietnamese physicist from the University of Melbourne, focuses mainly on data analysis for heavy-ion collisions at the [ATLAS experiment](#). She's also involved in the operation of the ATLAS detector.

Joni is passionate about science communication and education activities, especially for the young generation. Her passion for science was triggered by her curiosity to explore – in her own words – "worlds that are physically unreachable and invisible to the naked eye, like atoms and subatomic particles".

Joni shared with us a glimpse of one of her first big moments at CERN: "When I started working as a run control shifter, I was very nervous, but the shift leader, Clara Nellist, was very kind and supportive of the whole crew. Now that I've

become a shift leader myself, I'm truly grateful to Clara and everyone I've had a chance to work with at CERN, who constantly encouraged me to move beyond my comfort zone."

Livia, post-doc at the ALICE experiment

Livia, an Italian physicist, oversees the operation of the muon spectrometer of the [ALICE experiment](#). She's also doing research and development on silicon detectors for the upcoming ALICE detector upgrade.

Since secondary school, Livia has always been enthusiastic about science. The experiments in her school's laboratory and her passion for research convinced Livia to pursue a STEM career.

When we asked Livia about her favourite moments at work, she didn't hesitate: "The amazing and fun time I spent in the ALICE control room, waiting for the LHC beam to arrive, while preparing the detectors for data taking. The most fun for me is doing R&D on particle physics detectors, building them from scratch and then seeing them installed in the experiment caverns."

Tetiana, physicist at the ATLAS experiment

Tetiana is a Ukrainian physicist from the Annecy Particle Physics Laboratory in France, working on the [ATLAS experiment](#). She works on the upgrade of the electronics for an ATLAS calorimeter. She's also searching for phenomena beyond the Standard Model.

When we asked what inspired her to pursue a STEM career, Tetiana told us that it was an obvious choice for her, as everyone in her family was either a scientist or an engineer. "I decided to do a PhD in physics and mathematics when I was 10 years old."

Her favourite memory involving science from her childhood is growing "beautiful blue crystals from copper sulphate. They were growing on the

kitchen windowsill in my home in Kharkiv next to jars of green onions."

Deepti, engineer for the North Area Consolidation project

Deepti is an Indian engineer working on the project to consolidate CERN's North Area experiment facility.

Deepti was always intrigued by the basic principles of science and their everyday utility. Her affinity for STEM kept growing over the years and she decided to pursue a career in mechanical engineering.

"During my childhood, I was fascinated by buoyancy, gravity, density and water displacement as I watched paper boats float on water. When I was a young child, I learned to make paper boats and put them on running water during the monsoon season in India."

Alicia, PhD student in the accelerator field

Alicia is a Spanish doctoral student working on developing an ultra-fast generator for special magnets used in CERN's accelerators.

During her engineering studies, she enjoyed being in the lab the most, which inspired her to choose this path.

"When I was still living in Madrid, I used to go to a secondary school that was very close to the Residencia de Estudiantes [student housing] and I loved that place: the buildings are beautiful, the garden that surrounds it, everything. And then I read a discreet sign that said that this was also a historical site of the European Physical Society. Marie Curie had been there, Einstein too... That may have had an influence on my choice of a degree!" says Alicia.

Bianca Moisa

Record participation in the 2024 “Women and Girls in Science and Technology” event!

From 5 to 9 February, more than eighty female ambassadors from CERN, the University of Geneva, EPFL and LAPP spoke to 5800 local pupils, giving over 260 presentations in the space of a week! Their goal: to get children excited about science and break down gender stereotypes about scientific jobs.

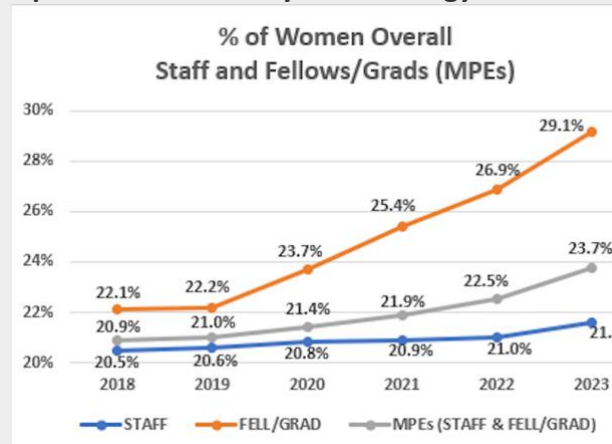
The Women and Girls in Science and Technology event has been an annual fixture since 2017, marking the International Day of Women and Girls in Science, which is celebrated on 11 February. The event’s eight editions have met with increasing success, thanks to an ever-expanding cohort of ambassadors eager to share their passion. In total, more than 24 500 pupils aged between 7 and 15 from the local region have seen for themselves that careers in science, technology, engineering and mathematics are equally accessible to girls and boys.

Are you a teacher who would like to take part in the 2025 event? Sign up for our education newsletter at <https://voisins.cern/en/newsletter-educational-offers> to find out what we offer and when registration will be open!

Would you like to take part in the 2025 event as a volunteer? Contact the CERN events team at education.locale@cern.ch to find out about our upcoming calls for volunteers !

More information on **CERN's Women in Technology group** on: <https://wit-hub.web.cern.ch/>.

Update on the “25 by ’25” strategy



(Image: CERN)

In spring 2021, the [Diversity & Inclusion programme](#) launched the “25 by ’25” strategy, an aspirational target-based initiative to boost the gender and nationality diversity of CERN’s staff and fellows population (MPEs) by the end of 2025. Objective: to reach 25% of women among MPEs by the end of 2025.

The latest statistics (see graphic) are very encouraging: CERN is only 1.3% away from its target!

Mélissa Samson

Introducing CERN’s robodog

A four-legged robotic solution has completed its first successful radiation protection test inside CERN’s largest experimental area

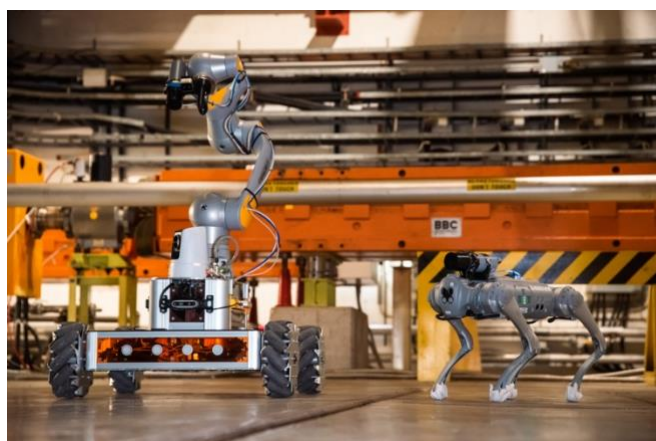
Building 937 houses the coolest robots at CERN. This is where the action happens to build and programme robots that can tackle the unconventional challenges presented by the Laboratory’s unique facilities. Recently, a new type

of robodog has entered CERN’s robot pool and successfully completed its first radiation protection test in the North Area.

“There are large bundles of loose wires and pipes on the ground that slip and move, making them

unpassable for wheeled robots and difficult even for humans. We carried out a proof-of-concept survey with the Radiation Protection group in this area. There were no issues at all: the robot was completely stable throughout the inspection,” said Chris McGreavy, a robotics engineer in CERN’s Controls, Electronics and Mechatronics (CEM) group.

Until today, CERN’s family tree of robots included the modular CERNbot in different sizes and configurations, such as the CERNbotSPS, as well as the Train Inspection Monorail (TIM) and CRANEbot. They can carry heavy payloads like robotic arms and other tools but are limited when it comes to entering cluttered areas and moving over unstructured surfaces and on steps.



(Right to left) Robodog with its counterpart, CERNBotNA – where NA stands for North Area. Together, these robots have completed a successful radiation protection survey inside CERN’s largest experiment area. (Image: M.Struik/CERN)

The team is now developing tools and advanced control algorithms for the robodog and its successors for long-term deployment in the experiment caverns, such as that of the ALICE detector, which are complex environments with metal stairs and narrow corridors designed for humans or, well, robots with legs. The current robodog is a commercially available product which CERN plans to use to explore possibilities to develop its own robotic solution in the future. In collaboration with the Experimental Physics R&D department, the CEM group is developing more

customised solutions based on four-legged robot that will soon be able to manoeuvre throughout almost the entire cavern. These robodogs will be able to monitor the state of the caverns and their environmental conditions regularly. They can identify water or fire leaks and other incidents, such as false alarms, in a timely manner, all of which can significantly impact the operation of the machines in the caverns and tunnels.

Each robot developed at CERN is carefully crafted to meet unique challenges and to complement each other. For example, there are rails attached to the ceiling running along the 27-kilometre tunnel of the Large Hadron Collider (LHC). The TIM monorail robot uses these rails to move around the tunnel. While it is great for monitoring and interacting with the tunnels from above, CERN’s new small robodog can perform activities on the ground, especially under the beamline, where no robot could tread before so easily. It is envisaged to be integrated with the four monorail robots currently in operation in the LHC.

“The TIMs are used for monitoring the large distances of the LHC from above and can travel long distances without recharging. They can deploy the quadbots in local areas to get more information about specific places that the TIM cannot easily access,” explains McGreavy.

The robodog will be able to enter new dimensions of the caverns, unlike the previous wheeled, tracked or monorail robots – expanding the range of environments that CERN robots can navigate. The Beams department continues to dream up robots for CERN and engineer them into reality. At the end of December 2023, the [LHCb](#) experiment [released](#) all its data from Run 1 of the Large Hadron Collider. This data, collected by the experiment in 2011 and 2012, contains approximately 800 terabytes of information obtained from proton–proton collisions. The data has been made available in a pre-filtered format, suitable for a wide range of physics studies for research and education purposes.

Chetna Krishna

HiLumi News: cool kickers for the HL-LHC

Successful tests of the first “MKI-Cool” show that these new kicker magnets can take the heat and keep their cool with high-luminosity beams

Kicker magnets are an important part of the LHC accelerator complex. Installed at the intersection of the LHC ring and the SPS transfer lines, they give each injected beam a “kick” at the right time to put it into orbit in the LHC.

The higher luminosity of the HL-LHC will pose a challenge for these magnets, as increased heat load could result in a miskick of the injected beam. To avoid this, engineers in CERN’s Systems department have developed a new version of the kicker magnet for the HL-LHC, called an “MKI-Cool”. One such magnet was installed in the LHC one year ago, replacing a standard kicker magnet. Measurements during its first year of operation, with high-intensity beam, show that the temperature rise of the MKI-Cool is less than one-fifth of that of the other seven kicker magnets in the LHC. This confirms that no heating issues should occur for the MKI-Cool kicker magnets with HL-LHC beams.

“Based on this excellent result, all the kickers will be sequentially upgraded to MKI-Cools,” says Mike Barnes, senior engineer in CERN’s Systems department. “The full upgrade of the MKIs will be completed during Long Shutdown 3.”*

Unlike other magnets in the accelerator, kicker magnets cannot be fully shielded from the beam. Shielding would interfere with the fast magnetic field pulse that they provide to kick the beam. In addition, the high-voltage pulse required prevents the magnets from being water-cooled, which is a serious hurdle as the ferrite they are made from loses its magnetic properties above the temperature of 125 °C. Under these conditions, the MKIs would miskick the injected beams, causing the downstream magnets to lose their superconductivity.

Following years of research and development, the team came up with a new design. The MKI-Cool design works by moving most of the beam-induced heating from the ferrite yoke to a so-called RF damper, which contains a ferrite cylinder and is mounted just upstream of the magnet. The beam-induced heat is then removed from the RF damper using a water-cooling circuit.

“The concept of moving the heat was demonstrated in computer simulations, but it was very challenging to prove this in lab-based measurements,” Barnes continues. “Hence, to fully prove the concept, a prototype with an RF damper, which was not cooled, was installed in the LHC in 2018 and measurements of temperature, with circulating LHC beam, proved that the RF damper concept worked effectively.”

To measure the difference in the beam-induced heat load between the MKI-Cool and the old kicker magnets, the team used temperature sensors attached to a nearby metal side plate. It was not possible to directly measure the temperature of the ferrite because it is pulsed to a very high voltage during the beam injection.

“During 2023 LHC operation with the MKI, the measured temperature of both the RF damper and the side plate remained relatively low,” Barnes continues. “Based on these temperature measurements and simulations, no heating issues are expected for the MKI-Cool in the HL-LHC era.”

SY Department

**Mike Barnes retired at the end of 2023: Giorgia Favia is now responsible for the MKI kicker magnets and will oversee the full upgrade to MKI-Cools.*

Happy hundredth Herwig!

Join in a rousing chorus of “Happy Birthday” on Friday 1 March, as CERN celebrates the 100th

birthday of Herwig Schopper, CERN Director-General from 1981 to 1988.

Herwig has made landmark contributions to nuclear and particle physics and to related technologies. In his early career, he played a key role in shaping today's physics research landscape in Germany, establishing laboratories and institutions before going on to leadership roles at DESY and CERN.

After retirement, not content to rest on his laurels, Herwig embarked on a new career: as a science diplomat. In this capacity, he played a leading role in the establishment of the SESAME laboratory in Jordan, a synchrotron light facility for the Middle East and neighbouring regions.

Over his remarkable career, Herwig has rubbed shoulders with the giants of the field, counting

many as friends. Few have had the opportunity to witness the evolution of particle physics from such a privileged vantage point.

Now is your chance to hear this history first hand. On **Friday 1 March from 2 p.m. in the Main Auditorium**, current and former CERN directors, eminent scientists and Herwig himself will speak, before participants are invited to raise a glass at a drinks reception. Full details are available at the Indico link.

Register now to join the celebration of Herwig's life and achievements to date:
<https://indico.cern.ch/event/1366175/>

Science Gateway welcomes its 100 000th visitor

Since opening in October, the centre has welcomed an average of 1000 visitors per day



From left to right: Nicky Morren and Bavo Lens, with Patrick Geeraert, head of CERN Science Gateway. (Image: CERN)

On Tuesday, 6 February, CERN Science Gateway welcomed its 100 000th visitor.

Bavo Lens and Nicky Morren came from Hasselt to Geneva on a city break and said “visiting CERN is a must”.

“For me, as an engineer, it was great to be able to see high-tech machines like the Synchrocyclotron and ATLAS,” Lens said. “Congratulations to the guide who was able to explain the very complex material in understandable language. The reception building is very beautiful and offers wonderful exhibitions that explain how particle research works very clearly, even for those who are not gifted in science. We ended our visit in the

restaurant, where we enjoyed the vegetarian options!”

Since the opening of CERN Science Gateway on October 8 2023, an average of 1000 visitors per day have enjoyed this new facility. The centre offers activities for all ages, including inviting young visitors from five years old to play and “see the invisible” while building up an interest in and connection to science and technology.

Having reached this milestone, the Visits service would like to send a big “thank you” to all its active guides. None of this would have been possible without the enormous dedication of each and every one of them, volunteering day after day to ensure that our visitors have an inspiring experience.

For those who have not yet found the time to [become a guide](#): take the first step and become part of this new era of outreach and education at CERN. The first step is usually the biggest, but the team will be there to support you at every stage of the journey.

The CERN Science Gateway is supported through the [CERN & Society Foundation](#) and the generosity of its donors.

Computer Security: Bull**** Bingo

There are many mantras and claims floating around about cybersecurity. Some of them leave no room for doubt, like “defence in depth”, which suggests deploying protective means at every level of the hardware and software stack, or “KISS – keep it simple, stupid” to avoid over-complication and too many deviations from the “standard” cybersecurity system. Other, more unfortunate statements also hold true. For example, “convenient, cheap, secure – pick two” makes “secure” always the least attractive option, as it brings no immediate benefits. However, some other mantras and claims are simply not true. Plain wrong. Or, excuse my language, “bull****”.

Indeed, computer security is never straightforward. Often, there is no single solution, but a series of complementary solutions is needed, like how our xorlab ActiveGuard solution works together with the Microsoft SPAM filter. Often a holistic solution cannot be found, for example when the quick fix of having two-factor authentication (2FA) for the new CERN SSO was deployed, which meant that the old SSO was left to die, and the non-holistic solutions we are looking at for how to deploy 2FA to LXPLUS and Windows Terminal Servers in the future.

Generally, computer security requires the aforementioned “defence in depth”: individually, multiple protective layers, each with a defined (implementation) scope, a limited coverage and holes are insufficient. But together, they provide adequate overall protection to the Organization that is pragmatic, balanced and efficient. Combined, they keep the cybersecurity risks and threats to the Organization under control.

So, while we acknowledge that there is no single solution to “cybersecurity”, there are many wrong solutions. Wrong statements. Wrong mantras. Bull****. In order to give you an idea of what we mean, let’s play “Bull**** Bingo”. Below are 25 statements we have heard in the past about cybersecurity, best security practices and cybersecurity implementation, some even from esteemed colleagues. Can you spot where they went wrong?

The first three people to send the five true statements to Computer.Security@cern.ch will win a bottle of Coca-Cola, as well as a “Hawaiian” pizza from CERN’s Restaurant 2.

CERN Computer Security team

	A	B	C	D	E
1	There is no malware for Apple devices	Software from the Google Play Store is harmless	Security is everyone's responsibility	SSH on port 2222/tcp is more secure	SPAM and malware filtering is 100% effective
2	2FA is a big step forward for account protection	Emails from "@cern.ch" are legitimate	I'm personally not a target as I'm not interesting to attackers	Back-ups cannot be altered	I have nothing to hide
3	I would never fall for phishing	Only the link <i>behind</i> a text/QR code reveals its truth	CERN's technical network is secure	A password written on a post-it is a good idea	QR codes always link to legit sites
4	A (free) VPN service protects me	Password protection on my laptop protects its data	My browser's password manager is secure	CERN is not interesting to attackers	CERN's anti-malware software is free for you to download
5	Using "https" means the website is secure	CERN's outer perimeter firewall keeps all threats away	Cloud services cannot be hacked	Encryption is easy; key management is complicated	WiFi is always secure

Want to learn more about computer security incidents and issues at CERN? Read our monthly reports (https://cern.ch/security/reports/en/monthly_rep

[orts.shtml](#)). For more information, questions or advice, check out our website (<https://cern.ch/Computer.Security>) or contact us at Computer.Security@cern.ch.

Official news

Summer work for children of members of the personnel

During the period from 10 June to 6 September 2024 inclusive, there will be a limited number of jobs for summer work at CERN (normally unskilled work of routine nature), which will be made available to children of members of the personnel (i.e. anyone holding an employment or association

contract with the Organization). Candidates must be aged between 18 and 24 inclusive on the first day of the contract and must have insurance covering illness. The duration of all contracts will be 4 consecutive weeks and the allowance will be CHF 1577 for this period. Candidates should apply

via HR department's electronic recruitment system, SmartRecruiters: <https://smrtr.io/j3W95>. Completed application forms must be returned by 11 March 2024 at the latest. To allow as many people as possible to benefit, each child may participate in this programme only once.

The results of the selection will be available mid of May 2024.

For further information, please contact:
Virginie.Galvin@cern.ch – Tel. 72855
(Geraldine.Ballet@cern.ch – Tel. 74151)

HR department

Your Google Workspace safely managed at CERN

Many people at CERN use Google Docs for work-related activities. To provide a safe environment and avoid issues relating to data storage and ownership, CERN has purchased Google Workspace subscriptions. Thanks to this, it will now be possible to access CERN Google Docs, authenticating with the CERN Single Sign-On. Of course, users will still be able to maintain a separate, personal Google account, using a non-CERN email account. The IT department will offer support to CERN people willing to move documents from their personal Google Docs to the new CERN-managed Google Docs environment. Following an extensive pilot phase, the new CERN-managed "Google Workspace" service is now available to every employed member of the personnel (MPE) and associated member of the personnel (MPA). By default, the only functionality enabled for CERN-managed Google accounts is Google Docs, as the service focuses primarily on live document editing. For convenience, additional functionality (via "bundles") can be enabled through the [portal](#). See [here](#) for a description of the bundles.

In order to protect your data, CERN's Google accounts have [restrictions](#) in place, which may prevent some functionalities from being enabled; if you need functionalities not available in the

bundles for personal reasons, it is advisable to continue to use your personal account. However, if you need professional access to such functionalities, you can open a ticket via the [CERN Service Portal](#).

Obtaining a Google account in the new CERN-managed Workspace is easy, but the procedure is different depending on whether you use a @cern.ch address or a @gmail.com address to access your private Google space. In detail:

- People who have previously created a '@cern.ch' Google account [can transfer](#) this account into the CERN-managed Google Workspace.
- People who use a private account (typically '@gmail.com') for their CERN work or who do not yet have a Google account can create a CERN-managed Google account [using the portal](#). The two accounts will remain completely separate, so you may need to transfer documents from one to the other.

To learn more about the differences between managed and unmanaged Google accounts, please visit the [glossary page](#).

IT department

Joint Advisory Appeals Board

The Joint Advisory Appeals Board has examined an internal appeal lodged by a staff member against

the Director-General's decision, which concluded an arbitration procedure concerning the

methodology of calculation of the salary increase annual budget in the context of the 2022 MERIT Guidelines and the decisions to qualify their personal performance during the year of 2021 as “strong”.

This appeal was receivable only regarding the individual performance qualification decision and resulting performance reward.

In application of Article R VI 1.18 of the Staff Regulations, and in agreement with the appellant,

the final decision of the Director-General, dated 10 May 2023, and the report of the Board are being brought to the attention of the members of personnel.

These documents will be available from **12 February 2024 until 1 March 2024** via this link: <https://indico.cern.ch/event/1380072/>

HR department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

27 Feb 11:00 | Physics | Main Auditorium | EP Seminar | EN

[Recent Results from IceCube and a Glimpse of the Neutrino Astronomy's Future](#)

1 Mar 14:00 | Knowledge Sharing | Main Auditorium | Symposium | EN

[Herwig Schopper - a century in physics](#)

5 Mar 13:15 | Computing | CERN Science Gateway | OQI | EN

[Launch event of OQI \(Open Quantum Institute\)](#)

7 Mar 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN

[CERN70 public event: From particle physics to medicine](#) (more [CERN70](#) events)

8 Mar 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 9 - Intergenerational dynamics in the workplace](#)

20 Mar 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[Status and Prospects of the JUNO neutrino experiment](#)

Registration now open

25–27 Mar | Computing | CERN | CERN openlab | EN

[2024 CERN openlab Technical Workshop](#)

15 May | Knowledge Sharing | Lausanne | Famelab | EN, FR or DE

[Famelab Switzerland local heats](#), open to 18–35-year-olds

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN

[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

12–25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN

[2024 Asia-Europe-Pacific School of High-Energy Physics](#)

23 Jul–1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN

[Accelerators for healthcare? 10-day innovation challenge](#)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN

[Introduction to Accelerator Physics](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch.

New webpage on ethics at CERN

Ethics is essential, in particular in our workplace. It contributes to a positive and productive work environment whilst fostering trust among us, as CERN contributors, and also the trust of our stakeholders, such as Member States, contractors, and research institutions.

At CERN, our ethics-related framework aims to guide us to act in accordance with the Organization's values and the conduct expected of us, and to establish and maintain a respectful working environment.

How to find CERN's Code of conduct and get more information:

In order to make our ethics-related framework more easily accessible, the HR department has grouped all the relevant information and

resources under a single point of reference at <https://hr.web.cern.ch/ethics>. On this webpage, you can find official reference documents (Code of conduct, applicable rules, policies and procedures), as well as other useful information.

Whether you want to browse relevant learning courses on offer, have more information on the anti-harassment and the anti-fraud policies, or learn about CERN's response channels in case of inappropriate behaviour in the workplace, visit the new "Ethics at CERN" webpage.

For more information or questions about ethics at CERN, please contact info-codeofconduct@cern.ch.

HR department

Ombud's corner

Overcoming a deadlock? Try reappraising!

Janet* has just been informed by her management that she has to step down from her current job and responsibilities and start a new project in another unit, taking on a role that poses a real challenge but entails no managerial responsibilities. Because of her excellent work in her current position, because her managerial skills have been widely recognised and most of all because she was not involved in the decision, Janet is struggling to accept this change.

She keeps wondering why this has happened and is very concerned about the potential impact of this change on her career. She feels shocked, worried and depleted of all energy. As a result, she is not sleeping well and has lost motivation for work. She needs to find a way to cope with the new circumstances and regain peace of mind and confidence.

Let's have a look at what happens to our brains in such circumstances, how emotions take over and what we can do to regain a sense of control.

Our brains like **certainty** and **autonomy**. A sense of uncertainty about the future and a feeling of being out of control both generate strong limbic system responses: our emotions, in this case negative ones.

In the situation described, Janet is both facing uncertainty about her future and experiencing a relative loss of autonomy. Hence the negative emotions that take over and start affecting her health.

In such situations, we may be tempted to try to suppress our emotions and even deny them. This does not work and, on the contrary, reinforces the negative feelings.

Janet could try to **label her emotions** by asking herself: what is it exactly that I am feeling now? Indeed, labelling and recognising emotions is proven to reduce them as they kick in. And, contrary to common belief, it will not make Janet feel worse.

Another very powerful tool to face challenging situations is to **reappraise them**, i.e. change the interpretation that creates the negative feelings in the first place.

After the much recommended “Stop, take a deep breath and think”, there are several ways to reappraise a situation, which we will try to apply to Janet’s case:

Reinterpreting the situation

This is when we consciously decide that the situation is no longer a threat. In Janet’s case, she could decide to see the new job as an opportunity to demonstrate another set of competences and skills, all assets for her future career.

Normalising the situation

Change is always difficult; it is normal to feel uncertain and apprehensive when facing a new situation that we did not necessarily choose. When Janet consciously realises that what she feels is normal and even to be expected, she will reduce her emotional response to the new situation.

Reordering priorities

We all have our set of values, in a given order of priority, which supports us in life. Janet gives

priority to the well-being of her family. Still, having managerial responsibilities and being able to influence things at a high level are also very important to her. Next in her current ordering of values comes the excitement of tackling challenging projects. Maybe Janet, to cope with the situation, needs to reconsider the relative importance of the last two, at least for a period of time, in order to find satisfaction in her new job and reduce her negative emotions.

Repositioning the perspective

A very powerful way to reduce emotions and turn them from an “away” to a “toward” reaction is to look at a situation from a different angle. Janet could also consider that her competences are critical to make her new mission a success and see that the management has chosen to entrust her with this highly challenging project. By doing so, she would change her negative emotions into excitement and regain peace of mind.

In challenging situations, we can, to some extent, control our brain’s response. It requires first being aware of our emotional responses, labelling them and applying effective reappraisal techniques.

If you are facing a difficult situation and are overwhelmed by your emotional response to the challenges, why not come to the Ombud’s office and discuss it?

If you would like to know more about how the brain functions and how we can manage our emotions, I highly recommend the great book from Dr David Rock, *Your brain at work*.

*The name and situation are fictitious.

Laure Esteveny

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>. More information on the role of the CERN Ombud and how to contact her can be found at <https://ombud.web.cern.ch>.